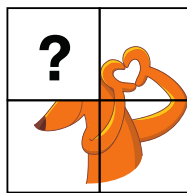


# CANADIAN MATH KANGAROO CONTEST PROBLEMS

## PART A: EACH CORRECT ANSWER IS WORTH 3 POINTS

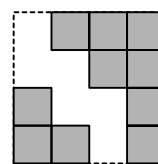
1. Which of these puzzle pieces completes the picture?



- (A)  (B)  (C)  (D)  (E) 

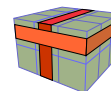
2. A large square is formed from 16 small squares of the same size. Then some of the small squares are removed. How many small squares are missing?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8



3. Two rubber bands are placed around a stack of 18 cubes. How many of the cubes are not touching a rubber band?

- (A) 6 (B) 8 (C) 9 (D) 10 (E) 12



4. There were three books on Mary's shelf, as shown. First, Mary swapped the white book and the grey book. Then Mary swapped the grey book and the black book. How are her books arranged now?



- (A)  (B)  (C)  (D)  (E) 

5. There were 12 pieces of fruit on the table. Vera removed 2 pears, 4 apples and half of the oranges. Now there are only oranges left on the table. How many oranges are left?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 6

6. Which rope ties into a knot when the ends are pulled?

- (A)  (B)  (C)  (D)  (E) 

7. Ann buys 3 chocolate bars. Tom buys 5 chocolate bars. Tom paid 8 euro more than Ann. What is the cost of 1 chocolate bar?

- (A) 1 euro (B) 2 euro (C) 3 euro (D) 4 euro (E) 5 euro

8. Charles has 5 wooden toys, as shown in the figure. He selects 2 pairs of toys with equal weight. Which toy did he leave out?

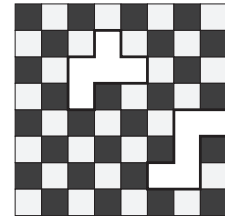
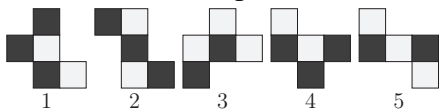


- (A)  (B)  (C)  (D)  (E) 

9. 3 years ago, the sum of Ana's age and Ben's age was 6 years. Ana is currently 7 years old. How old is Ben now?

- (A) 1 year (B) 5 years (C) 6 years (D) 7 years (E) 11 years

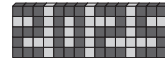
10. Which two of the pieces shown below complete the chessboard?

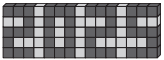
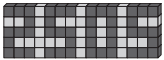
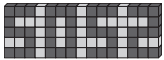
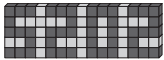
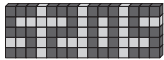


- (A) 1 and 2 (B) 1 and 5 (C) 3 and 4 (D) 3 and 5 (E) 4 and 5

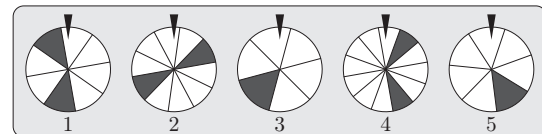
## PART B: EACH CORRECT ANSWER IS WORTH 4 POINTS

11. Anna has built a wall that displays the year 2025. Bella stands on the other side of the wall. What does Bella see?



- (A)  (B)  (C)  (D)  (E) 

12. The figure shows five wheels of fortune. Each wheel is divided into a different number of identical parts. You will win a prize when the wheel is spun and then stops with the triangle above the wheel pointing to a part that is shaded. Which wheel gives you the best chance of winning?



- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

13. Lisa has four wooden digits. She can use them to form the number 2025. Which of the following numbers is the largest she can form with these digits?



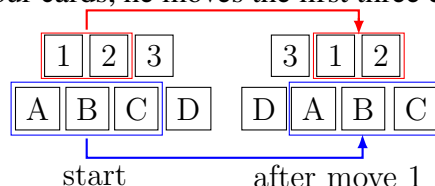
- (A) 2502 (B) 5202 (C) 5220 (D) 5502 (E) 5520

14. In the petting zoo, Renée feeds 6 sheep. She gives them a total of 210 grams of dry food for lunch. She gives the smallest sheep twice as much food as she gives to each of the others. How much does the smallest sheep get?



- (A) 55 grams (B) 60 grams (C) 70 grams (D) 75 grams (E) 80 grams

15. Joe mixes two rows of cards as follows: In the row with three cards, he moves the first two cards to the end. In the row with four cards, he moves the first three cards to the end.

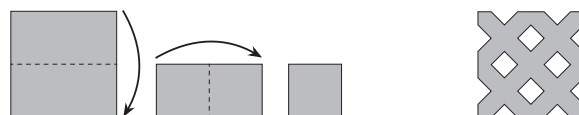


After how many moves will card B and card 2 both be in the first position at the same time for the first time?

- (A) 11                      (B) 12                      (C) 6                      (D) 15                      (E) 9

16. Nela folds a paper square in half and then in half again, as shown.

Next she cuts pieces out of the folded paper.  
After unfolding she sees a paper snowflake.



How did she cut the folded piece of paper?

- (A)                       (B)                       (C)                       (D)                       (E) 

17. Mary and Paul each wrote down three 3-digit numbers using the digits 1 to 9 exactly once. Then they ordered their numbers as smallest, middle and largest. Mary wrote down the largest possible value the middle number could have.



Paul wrote down the smallest possible value the middle number could have.

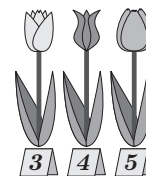
What is the difference between their two middle numbers?

- (A) 642                      (B) 684                      (C) 864                      (D) 888  
(E) none of the previous

18. Zeta wants to buy some flowers.

The prices of the three flowers she can buy are shown in the figure.

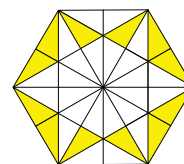
How many different bouquets with a total cost of exactly 23 can she buy?



- (A) 4                      (B) 5                      (C) 6                      (D) 7                      (E) 8

19. The regular hexagon shown is divided into many triangles of equal area.

Which fraction of the hexagon is shaded?



- (A)  $\frac{1}{2}$                       (B)  $\frac{1}{3}$                       (C)  $\frac{1}{4}$                       (D)  $\frac{1}{5}$                       (E)  $\frac{1}{6}$



20. Ohad wants to write the four digits 2, 0, 2 and 5 in the four boxes of the calculation shown. What is the smallest result that Ohad could get?  $\square - \square + \square - \square$
- (A)  $-7$       (B)  $-6$       (C)  $-5$       (D)  $-4$       (E)  $-3$

**PART C: EACH CORRECT ANSWER IS WORTH 5 POINTS**

21. Kate and Tom are celebrating their birthday today. Tom notices that  $\frac{1}{19}$  of Kate's age is equal to  $\frac{1}{17}$  of his age. The sum of their ages is greater than 40 and less than 100. How old is Kate?

(A) 19      (B) 34      (C) 38      (D) 57      (E) 76

22. The leaflet shown includes transparent windows, allowing what is below to be clearly seen when the flaps are folded over. When both flaps are folded over, what is the sum of the numbers that can be seen through the windows?

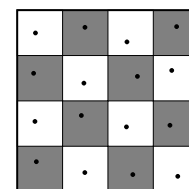
|  |  |   |   |   |  |  |
|--|--|---|---|---|--|--|
|  |  | 4 | 9 | 2 |  |  |
|  |  | 3 | 5 | 7 |  |  |
|  |  | 8 | 1 | 6 |  |  |

(A) 7      (B) 9      (C) 12      (D) 14      (E) 15

23. Mike obtains a number  $x$  by dividing the number  $\sqrt{11}$  by three. Where is the number  $x$  to be found on the number line?

(A) Between 0 and 1      (B) Between 1 and 2      (C) Between 2 and 3  
(D) Between 3 and 4      (E) Between 4 and 5

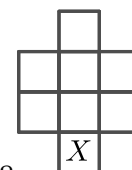
24. On a  $4 \times 4$  giant chessboard there are 16 kangaroos, one in each square. On each turn, each of the kangaroos jumps to a neighbouring square (up, down, left or right, but not diagonally). All kangaroos stay on the board. There can be several kangaroos on any square.



After 100 turns, what is the largest possible number of empty squares?

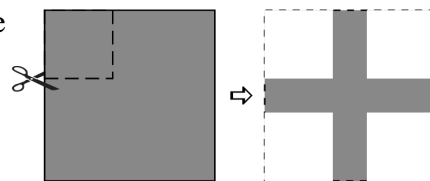
(A) 15      (B) 14      (C) 12      (D) 10      (E) 8

25. David wants to place the numbers 1 to 8 in the eight cells of the diagram, with one number in each cell. He wants the cells that contain two consecutive numbers not to share a side or a vertex. Which numbers can David put in cell marked X?



(A) 1 or 8      (B) 2 or 7      (C) 3 or 6      (D) 4 or 5      (E) 7 or 8

26. Janaína cuts four identical squares from the corners of a square sheet of paper, as shown. The total area she cut off is  $16 \text{ cm}^2$  and the area of the cross that remains is  $9 \text{ cm}^2$ .

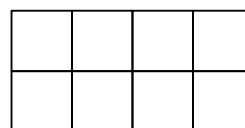


What is the perimeter of the cross in cm?

- (A) 9                      (B) 16                      (C) 20                      (D) 25                      (E) 32
27. When grandma started knitting woollen socks, she had a huge ball of yarn with a diameter of 30 cm. After finishing 70 socks, she still has a ball of yarn with a diameter of 15 cm. How many more socks can grandma knit with the yarn that is left?

(A) 70                      (B) 50                      (C) 30                      (D) 20                      (E) 10

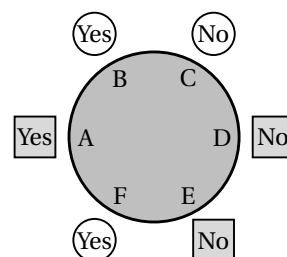
28. Anastasia wants to write the numbers from 1 to 8 into the cells of a  $2 \times 4$  grid. The number in each cell must be smaller than the number in the cell to its right and smaller than the number in the cell below it.



In how many different ways can Anastasia fill the grid?

- (A) 6                      (B) 8                      (C) 10                      (D) 12                      (E) 14

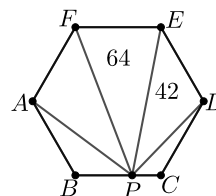
29. A group of three square men from Mars and a group of three circular men from Jupiter are sitting around a table, as shown. One of the six has the key to their flying saucer. All members of one group always tell the truth and all members of the other group always lie. All six were asked the question “does a person sitting next to you have the key?” Their answers are shown in the figure.



Who has the key?

- (A) A                      (B) B                      (C) C                      (D) D                      (E) E

30. The diagram shows a regular hexagon  $ABCDEF$ . Point  $P$  lies on  $BC$  so that the area of  $\triangle PEF$  is 64 and the area of  $\triangle PDE$  is 42.



What is the area of  $\triangle APF$ ?

- (A) 53                      (B) 54                      (C) 56                      (D) 60                      (E) 64

**Answer Keys:** ECADCEDDBB BACBABADBC CDBBBCEEBB